

Peter Hinchley

SOLUTION ARCHITECT



I have over twenty years' experience working as a Systems Engineer and Solution Architect. I have designed and deployed desktop and server standard operating environments for several government departments. I have a deep technical knowledge of the Windows operating system and infrastructure services including Microsoft Configuration Manager and Active Directory. I have strong application development and packaging skills, and experience in both on-premise and cloud-based delivery models. I have also designed and deployed multiple large-scale projects, including a global privilege management service for BHP Billiton.

CONTACT

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CLEARANCE:
NV1 (Secret)

EDUCATION

BACHELOR OF COMPUTER
ENGINEERING

First Class Honours
University of Canberra [94 – 97]

HIGH SCHOOL CERTIFICATE

Dux of School
Narrandera [1988 – 93]

SKILLS

Desktop and Server Standard
Operating Environments

Application Packaging

Systems Integration

Identity and Access Management

Programming / Web Development

Cloud Engineering / Cyber Security

Solution Architecture

CLOUD ENGINEER.

ACCC (2024 - 2024).

Worked to commission, enhance, and support the cloud infrastructure underpinning the National Anti-Scam Centre. This involved working with a broad collection of cloud resources including Azure DevOps, Azure API Management, Azure B2C, Azure Front Door, Azure Web Application Firewall, Azure Data Factory, Azure SQL, Azure Machine Learning, Azure App Service, storage accounts, key vaults, private endpoints, etc.

Also provided support to several other large scale cloud initiatives run by the ACCC, including Gas and Supermarket Inquiries and Merger Reform.

Designed and implemented an Infrastructure as Code (IaC) framework for the agency. This framework was based on Bicep, and leveraged a library of reusable custom templates. It made use of deployment stacks for clearly differentiating between managed and unmanaged resources. It also incorporated several advanced techniques, such as automated module testing, with support for custom linting rules, and automated budget creation.

Implemented a mechanism, based on API Ops, to migrate Azure API Management configuration between environments using automated pipeline deployments.

Developed a policy document governing the creation of all resources in Azure. This included guidance for the creation of management groups and subscriptions, and naming requirements based on resource type. The latter convention was supporting using regular expression definitions that were enforced using custom PSRule policies.

Designed and implemented a system for assigning budgets with associated cost notification alerts across all Azure resource groups. Also supported this work by developing an automated process for generating detailed historical cost expenditure reports (in Power BI) based on custom tag assignments.

Worked on the design and build of two separate Azure DevTest Labs solutions. The first allowed developers to build personal development workstations based on a collection of custom and marketplace images. The second was a sandboxed solution that enabled secure manual inspection of potential scam sites using ephemeral Linux machines that could be provisioned on demand. The latter solution was entirely built using IaC. It leveraged pipelines for automating the maintenance of a custom Linux image, Azure Bastion for mediating access, and Azure Data Share for managing the export of data out of the sandboxed environment.

Worked closely with developers in an agile operating model to quickly provision cloud resources and access as required, and to investigate, diagnose, and resolve complex third level deployment and integration issues.

CLOUD TRANSITION / SOLUTION ARCHITECTURE.

Comcare (2018 - 2024).

Identity and Access Management

Designed and implemented an identity synchronisation and provisioning solution based on Microsoft Identity Manager. The system links Aurion (HR), Active Directory, and the building access system. It is used to provision new starters in Active Directory, automatically suspend and retire the accounts of staff that are on leave, or whom have left the agency, and to flow identity and organisational changes between systems.

Implemented Okta - a cloud-based identity and access management solution - for managing the identities and entitlements of staff and external users. The solution enables user self-service capabilities, multi-factor authentication, delegated administration, and single sign-on across corporate and cloud-based applications.

Integrated Okta with a broad range of IT systems including Office 365, the corporate intranet (Squiz), Wellnomics, Aurion, Technology One, Citrix NetScaler, Cisco Jabber and WebEx, Acorn (LMS), a Dynamics 365 Portal, and an in-house developed Angular application and backend REST API. These integrations cover technologies including OpenID Connect, OAuth 2.0, SAML, and WS-Federation.

Developed a custom SCIM 2.0 compliant REST API (coded from scratch using Node.js) that facilitated synchronisation of identities between Okta and the Comcare Claims Online system.

Established an on-premise open-source identity service to facilitate the development of custom applications that rely on OpenID Connect (without dependency on cloud resources).

Upgraded the existing on-premise Microsoft Active Directory Federation Services (ADFS) infrastructure, and then subsequently migrated all systems federated with ADFS to Okta.

DevOps and Cloud Services

Upgraded and then migrated the on-premise Microsoft TFS solution to Microsoft DevOps. This included migrating existing TFVC source code repositories to Git.

Established automated build and deployment pipelines for several in-house developed applications (builds are configured in TeamCity and deployments in Octopus). Also provided best practice guidance and implementation support for colleagues seeking to adopt DevOps practices.

Proposed, designed, and implemented an automated solution for managing the maintenance of SSL certificates using Let's Encrypt. This included writing a PowerShell script for assigning and updating certificates on public-facing NetScaler infrastructure.

Developed architectural guidance and implementation procedures enabling developers to leverage Docker containers as part of the software development lifecycle. This work also bootstrapped a recent trial of an on-premise Azure API Management gateway (hosted on VMware Integrated Containers).

Implemented "co-management" between Microsoft Configuration Manager and Intune and configured Intune conditional access policies for managing access to Office 365 resources from both corporate and personal devices.

Developed and tested a procedure for migrating user home drives to Microsoft OneDrive. This included working around an existing folder redirection policy that was incompatible with OneDrive.

Systems Integration

Designed and implemented an API proxy based on an open-source platform named Tree Gateway. The solution provides key functions including dynamic routing, request throttling, a fail-fast circuit breaker, response caching, OAuth token

validation, and the capability to dynamically modify both request and response headers and body content. The solution is written in TypeScript and leverages a clustered Redis service for persisting logs and system configuration data.

Established an on-premise log aggregation service using both Windows event-forwarding and NxLog agents to forward key security events to a cloud-based SIEM based on FireEye Helix.

Upgraded the solution used to automate the conversion of emails and Office documents to PDF format (based on Adlib Enterprise). This included writing custom integration code for interfacing to Microsoft Exchange and developing a process to dynamically resize inline attached images to fit within designated page dimensions.

Implemented significant improvements to the security and data quality of the corporate Active Directory environment. This included the identification and removal of obsolete/redundant artefacts such as security groups, service accounts, and group policies. It also included the introduction of technical controls for the use of domain administrator accounts (based on Microsoft authentication silos). As part of this work, the use of multi-factor authentication was enforced for privileged activities performed via designated administration hosts.

Developed numerous PowerShell scripts to automate critical business processes. This included developing multiple add-ons to Microsoft Outlook. One was used to generate a consolidate PDF of emails (including nested attachments), and another was used to perform a bulk upload of emails to HPE Content Manager. Also made extensive use of PowerShell DSC to facilitate the declarative configuration of server infrastructure.

Architecture and Governance

Developed a long-term strategy for the modernisation of the end-user computing environment. The blueprint outlined a roadmap for migrating from the on-premise device management product suite (based on Microsoft Deployment Toolkit, Configuration Manager and group policy) to a pure cloud-only toolchain (using technologies such as Okta, Microsoft AutoPilot and Intune).

Published several architectural and corporate standards documents, including an HTTP REST implementation policy, a comprehensive suite of identity and access management principles (covering technologies such as OAuth 2.0 and OpenID Connect), and defining the agency's register of officially supported web browsers.

Formulated a role-based access control model for allocating resource privileges to IT support staff.

Participated in a range of tender evaluation panels for services including enterprise architecture, identity and access management, claims payment automation, network refresh, API management, and a Dynamics 365 integration partner.

Implemented a web portal used by the IT Service desk for performing common administration tasks. The portal provides a streamlined graphical interface to a collection of PowerShell scripts that perform complex backend administrative tasks in a consistent, secure, and repeatable manner.

DESKTOP SOE / APPLICATIONS PACKAGING.

Comcare (July 2017 - 2018).

I was the sole technical resource on the design, build, and deployment of a Windows 10 SOE (1703 and 1709). The reference image was developed using Microsoft Deployment Toolkit and deployed via Configuration Manager. The project scope included the configuration of DirectAccess, BitLocker, Endpoint Protection, and the deployment of Office 365. Additional tasks included the configuration of group policy, application whitelisting, the design of the ongoing servicing model, and the packaging of over 180 applications.

SERVER STANDARD OPERATING ENVIRONMENT.

Department of Border Protection (June 2017).

In this short-term contract, I designed and built a Windows Server 2016 SOE for the End User Computing Program. The reference image was built and deployed via Configuration Manager and the configuration of server roles was facilitated using PowerShell Desired State Configuration (DSC) policies. The pre-staging of servers in Configuration Manager was enabled through a purpose-built web user interface that invoked a custom PowerShell web service.

DESKTOP SPECIALIST.

Department of Foreign Affairs and Trade (Jan 2013 - May 2017).

I was the technical lead on the development of a Windows 10 SOE that was deployed to 10,000 computers in over 100 countries. I also developed the Windows 2012 R2 SOE and a Microsoft SCCM 2012 R2 solution (with approximately 500 site system servers). Other achievements included designing a custom Linux thin-client image with support for both RDP and ICA connectivity. I also led the development of a briefcase solution for providing staff with mobile access to the Department's SECRET network.

DIRECTOR DESKTOP AND ACCESS SERVICES.

Department of Finance (2010 - May 2012).

I was responsible for the strategic direction, implementation and operational support of the following services: Desktop, Directory Services, Security and Mobility. I led the project to upgrade Active Directory 2003 to 2008 R2 and designed and built the authentication platform used by the Central Budget Management System and Online HR solution for Parliamentarians. I also took a lead role in delivering the Department's remote access solution, web proxy, and printer consolidation projects.

DESKTOP SOE / APPLICATIONS PACKAGING.

Department of Finance (2008 - 2010).

I was the lead engineer responsible for delivering a Microsoft Windows Vista SOE for the Department of Finance. Key tasks included: configuring Microsoft MDT 2008 and SCCM 2007, packaging over 160 applications, defining group policy, and establishing a test framework to validate the implementation. After the deployment, I was responsible for third level support of the platform, and the delivery of ongoing performance and security improvements.

JOINT OPERATIONS PORTAL.

CSC (2007 - 2008).

In the role of Portal Architect, I lead a small team tasked with the design and development of a national Portal for the Department of Defence. The Portal was designed to support the operational planning and execution activities undertaken by all three divisions of the Australian Defence Force. The solution was based on Microsoft Office SharePoint 2007.

ENTERPRISE META-DIRECTORY.

CSC (2007).

Designed, developed, and deployed, an enterprise meta-directory for BHP Billiton that synchronised employee, contractor, and vendor identity data for over 120,000 objects between multiple SAP systems, Active Directory and AD LDS. The solution was built using Microsoft Identity Integration Server.

ENTERPRISE WEB ACCESS MANAGEMENT.

CSC (2002 - 2007).

Designed, developed, and deployed, a global web access management service for BHP Billiton. The system was deployed in five continents and protected web sites hosted across over 80 servers. The solution was the "corporate standard" authentication and authorisation service for all new web applications in the enterprise. The service processed approximately 1 million authentication events each month, supported a range of multi-factor authentication options, and included deep integration into Documentum and Outlook Web Access.

SELF-SERVICE PASSWORD RESET.

CSC (2006).

Designed, developed, and deployed, a global self-service password reset solution for BHP Billiton supporting approximately 60,000 users. The solution was based on CA Identity Manager.

EXTERNAL USER DIRECTORY.

CSC (2005 - 2007).

Was responsible for the design and delivery of an authoritative user repository for managing identities external to the organisation (vendors, suppliers, selected contractors, etc). The user store was replicated across five continents. As part of this work I developed a custom ASP application enabling delegated management of external users by authorised personnel.

DIRECTORY ARCHITECTURE.

CSC (2004 - 2005).

Prepared an identity management framework for the Department of Immigration that addressed functional requirements including synchronisation, provisioning, password management, access control, single sign-on, and reporting. I also designed a directory service capable of supporting the eBusiness initiatives of the department.

WEB HOSTING.

CSC (2002 - 2003).

Assisted in the design and deployment of a global web hosting service for BHP Billiton. The facility provided a three-tier architecture and was deployed on Solaris hardware. It supported high profile web sites including bhpbilliton.com and bmacoal.com. Key tasks included developing the logical technology model for the internet and intranet hosting zones, configuring all application software, and establishing the code deployment methodology and toolset.

ENTERPRISE META-DIRECTORY (SIEMENS).

CSC (2001 - 2002).

Assisted Siemens in the development and deployment of an enterprise meta-directory that synchronised employee, contractor, and vendor identity data for over 60,000 objects between SAP HR, Windows NT and Active Directory. The solution was built using Siemens DirX.

WEB HOSTING.

Conde Nast (2000 - 2001).

Implemented significant improvements to the infrastructure hosting Conde Nast (UK) web sites including Vogue, Traveller, Brides, and GQ. This included the introduction of load balancing and clustering technology to remove existing single points of failure.

ASNET AND CABNET.

CSC (1998 - 2000).

The lead architect in the design and implementation of the Australian Secure Network (ASNET) which serviced counter terrorist and dignitary protection activities, and CABNET, which enabled the secure electronic dissemination of Cabinet documents produced by Federal Parliament. Both solutions were based on Windows NT and Lotus Domino.

TECHNOLOGIES

Azure / Microsoft 365
Windows Server / 11
Entra ID / Active Directory / Okta
Intune / Configuration Manager
Hyper-V / VMware / AVD
SQL Server / Exchange

LANGUAGES

PowerShell
JavaScript / TypeScript / Node.js
PHP
C# / VB.NET / VBScript
HTML / CSS
C / C++ / Go
